

Competitive Season-Long Changes in Countermovement Vertical Jump Force-Time Metrics in Female Volleyball Players

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Abstract

Cabarkapa, DV, Cabarkapa, D, Philipp, NM, and Fry, AC. Competitive season-long changes in countermovement vertical jump force-time metrics in female volleyball players. *J Strength Cond Res* 38(2): e72–e77, 2024—Although force plates remain one of the most widely used tools for neuromuscular performance assessment in applied sports-specific settings, there is still a lack of scientific literature focused on studying changes in countermovement vertical jump (CVJ) performance in team sports such as volleyball, especially within the female athlete population. Thus, the purpose of the present study was to examine season-long neuromuscular performance changes in volleyball players. Eighteen National Association of Intercollegiate Athletics Division-I collegiate female athletes performed 3 maximal-effort CVJs while standing on a uniaxial force plate system sampling at 1,000 Hz at 5 different testing timepoints throughout a competitive season span (~11 weeks). The testing sessions were separated 2–3 weeks apart and performed at the approximately same time of the day (12:00 hours). Repeated-measures analysis of variance revealed that both concentric and eccentric force-time metrics remain relatively unchanged throughout a regular season span (e.g., concentric peak force and power, eccentric impulse and duration). However, the eccentric metrics such as peak and mean power and peak velocity displayed a slight improvement after a brief tapering period purposely implemented before the post-season competition to optimize the athlete's recovery (~15, 18, and 14% increase, respectively). In addition, the outcome metrics such as vertical jump height and reactive strength index-modified did not display notable fluctuations across the competitive season span. These findings can help coaches, sports scientists, and strength and conditioning practitioners to obtain a deeper insight into collegiate female athletes' force-time characteristics that may aid with developing adequate training regimens targeted toward optimizing on-court performance.

Key Words: monitoring, team sports, force plate, training load, coaching, testing

Introduction

The vertical jump is one of the most popular non-invasive and time-efficient assessments for monitoring neuromuscular performance in athletes that can be implemented in both field- and laboratory-based settings (23,25). The two frequently used tests in a practical setting are the squat jump, also called the static jump, and the countermovement vertical jump (CVJ) (25). The squat jump is initiated from the semi-squat position and requires athletes to jump vertically as high as possible without any countermovement before the takeoff (35). However, the CVJ is initiated from the upright standing position, commonly performed with or without an arm swing. Unlike the squat jump, it includes a rapid eccentric action followed by a short transition (i.e., amortization phase) that leads into an explosive concentric action (34,35). This combination of eccentric and concentric muscle actions that occurs during CVJ is known as the stretch-shortening cycle and it has been shown to have a performance-enhancing effect when compared with the squat jump (3,21,35).

Over the years, various performance-monitoring technologies such as force plates, accelerometers, motion capture systems, contact mats, and optical measurement systems have been used

for monitoring the impact of neuromuscular fatigue on CVJ performance (4–7,27). Among them, force plates have often been considered a criterion measure or “gold standard” for neuromuscular performance assessment, primarily because of their high sampling frequency (e.g., 1,000 Hz) and ability to provide practitioners with a plethora of force-time metrics (e.g., eccentric mean force, countermovement depth, and contraction time) (2,6,24,26,27). This allows sports scientists and strength and conditioning practitioners to not only measure outcome metrics (e.g., vertical jump height) but also gain a thorough understanding of biomechanical characteristics and different movement strategies that take place throughout various phases of CVJ (i.e., eccentric and concentric phases) (27).

A considerable amount of scientific literature has examined the acute and chronic effects of training and the competitive season on CVJ performance using force plate technology (9–11,14,16,22,28). Interestingly, the findings are inconsistent, and they seem to vary based on the competitive level and sex. Some research reports observed a decrease in CVJ performance throughout the competitive season span in junior and senior national team female basketball players (22), whereas others found improvements in power, quickness, and reaction time (16). In a similar investigation focused on examining neuromuscular performance changes in National Collegiate Athletic Association (NCAA) Division-I male basketball players, Murr et al. (28)

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found a significant increase in vertical jump height pre-post competitive season (i.e., 44.5–47.7 cm). In particular, 8 of 9 players improved their vertical jump height over the course of 18 weeks, whereas 1 player maintained the same vertical jump height performance levels (28). However, it should be noted that the authors did not observe significant changes in the average relative propulsive power and reactive strength index-modified (RSI-modified) (28). Furthermore, a recently published study by Spyrou et al. (32) examined changes in kinetic and kinematic parameters (e.g., peak power and velocity, and rate of force development) obtained during both eccentric and concentric phases of the CVJ motion in elite futsal players over the entire competitive season. The findings revealed that the competitive period suppressed players' ability to improve jumping performance and power during the concentric phase of the CVJ (32). Also, it should be noted that previous research reports have indicated that starters in team sports (e.g., basketball) were more likely to maintain their physical performance capabilities throughout the competitive season than nonstarters, regardless of their sex and level of play (15,16).

Although force plates remain one of the most widely used tools for neuromuscular performance assessment in applied sports-specific settings, there is still a lack of scientific literature focused on studying changes in CVJ performance in team sports such as volleyball, especially within the female athlete population. Thus, to bridge a gap in the scientific literature, the purpose of the present study was to examine season-long neuromuscular performance changes in collegiate female volleyball players.

Methods

Experimental Approach to the Problem

The present study has used repeated-measures research design to investigate season-long neuromuscular performance changes in collegiate female volleyball players.

Subjects

Eighteen female volleyball athletes (age = 20.4 ± 1.0 years; body mass = 72.5 ± 12.7 kg; height = 176.2 ± 8.7 cm; playing experience = 8.8 ± 2.8 years) with >4 years of previous competitive experience (e.g., high school and community college) currently playing at the National Association of Intercollegiate Athletics Division-I level of competition volunteered to participate in the present investigation. All athletes were free of musculoskeletal injuries and were cleared by their respective sports medicine staff to participate in the team activities. The testing procedures were previously approved by University of Kansas's Institutional Review Board, and after being informed about benefits and risks pertaining to participation in the present investigation, all athletes signed the informed consent form.

Procedures

Before the start of the data collection (i.e., initial testing time-point), all athletes were thoroughly familiarized with the testing procedures, including the CVJ technique. Data were collected at 5 different testing time points throughout the competitive volleyball season (~11 weeks). The first CVJ assessment was conducted one week before the beginning of the official pre-season, followed by subsequent testing sessions approximately 2–3 weeks apart. The last CVJ assessment was performed one week before the

conference tournament (i.e., post-season), after a brief tapering period. A graphical representation of the testing timeline is presented in Figure 1.

Upon arrival at the volleyball facility, all athletes performed a standardized warm-up protocol consisting of dynamic stretching exercises (e.g., high-knees, butt kicks, A-skips, forward lunges, lateral lunges). After the warm-up completion, each athlete was instructed to step on a uniaxial force plate system (ForceDecks Max, VALD Performance, Brisbane) sampling at 1,000 Hz and perform 3 maximal-effort CVJ with no arm swing (i.e., hands on the hips throughout the entire movement). Each jump was separated by a 10–15 second rest interval to minimize the possible influence of fatigue (6,26,30). The athletes were verbally encouraged to give maximal effort and focus on pushing the ground explosively as possible (19). Identical CVJ testing procedures were carried out during each testing session and were conducted at the approximately same time of the day (i.e., 12:00 hours). In addition, it should be noted that all athletes regularly participated in team resistance training sessions 2–4 times per week. Both training sessions and testing protocols were administered and supervised by Certified Strength and Conditioning Specialists.

Based on the previously conducted scientific literature and research conducted in our laboratory, the following force-time variables were obtained from the force plate system: braking phase duration and impulse, eccentric and concentric duration, peak velocity, mean and peak force and power, countermovement depth, contraction time (i.e., eccentric and concentric time), vertical jump height, and RSI-modified (1,17,30). A detailed description of all dependent variables examined in the present study can be found in the VALD user manual (<https://valdperformance.com/forcedecks/>) and previously published research reports (17,30).

Statistical Analyses

Descriptive statistics, mean values and standard deviations ($\bar{x} \pm SD$), were calculated for each dependent variable examined in the present study. Shapiro-Wilk's test and Q-Q plots were used to test that the assumption of normality was not violated. Because of sport-specific constraints, 7.7% of the data were missing. This issue was addressed by using the linear regression imputation method where the missing data were replaced by the calculated values obtained from the regression equation created using the existing data (18). Repeated-measures analyses of variance (ANOVA) with Tukey post hoc comparisons were used to examine statistically significant differences in each dependent variable across 5 testing time points throughout a full competitive season. Cohen's d was used to calculate the effect sizes (i.e., $d = 0.2$ small effect, $d = 0.5$ moderate effect, and $d > 0.8$ large effect) (8). Statistical significance was set a priori to $p < 0.05$. All statistical analyses were completed with SPSS (Version 26.0; IBM Corp., Armonk, NY).

Results

Descriptive statistics for each dependent variable examined in this study are presented in Table 1. The repeated-measures ANOVAs revealed no statistically significant differences ($p > 0.05$) in the force-time metrics obtained during the concentric phase of the CVJ across 5 different testing time points. Specifically, there were no significant changes in concentric mean force ($F_{[4, 14]} = 1.171, p = 0.365, \Lambda = 0.749$), concentric peak force

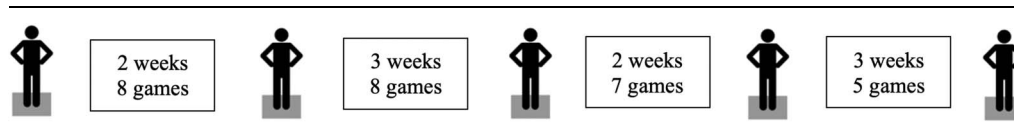


Figure 1. Graphical representation of the testing timeline. CVJ = countermovement vertical jump.

($F_{[4, 14]} = 0.573, p = 0.687, \Lambda = 0.859$), concentric mean power ($F_{[4, 14]} = 1.065, p = 0.410, \Lambda = 0.767$), concentric peak power ($F_{[4, 14]} = 1.183, p = 0.361, \Lambda = 0.747$), concentric peak velocity ($F_{[4, 14]} = 0.993, p = 0.443, \Lambda = 0.779$), concentric impulse ($F_{[4, 14]} = 1.727, p = 0.200, \Lambda = 0.670$), and concentric duration ($F_{[4, 14]} = 1.563, p = 0.238, \Lambda = 0.691$). Additionally, no significant changes were observed in RSI-modified ($F_{[4, 14]} = 1.153, p = 0.373, \Lambda = 0.752$), contraction time ($F_{[4, 14]} = 0.222, p = 0.922, \Lambda = 0.940$), vertical jump height ($F_{[4, 14]} = 1.085, p = 0.401, \Lambda = 0.763$), and countermovement depth ($F_{[4, 14]} = 2.162, p = 0.443, \Lambda = 0.779$).

However, several eccentric variables demonstrated significant alterations throughout the competitive season. Eccentric mean power ($F_{[4, 14]} = 3.935, p = 0.024, \Lambda = 0.471$), eccentric peak power ($F_{[4, 14]} = 3.835, p = 0.026, \Lambda = 0.477$), eccentric peak velocity ($F_{[4, 14]} = 3.028, p = 0.045, \Lambda = 0.536$), and eccentric mean force ($F_{[4, 14]} = 3.523, p = 0.034, \Lambda = 0.498$) revealed significant changes across the 5 testing sessions, whereas no changes were observed in eccentric peak force ($F_{[4, 14]} = 0.510, p = 0.730, \Lambda = 0.873$), eccentric duration ($F_{[4, 14]} = 0.937, p = 0.471, \Lambda = 0.789$), eccentric braking impulse ($F_{[4, 14]} = 1.097, p = 0.396, \Lambda = 0.761$), or braking phase duration ($F_{[4, 14]} = 1.153, p = 0.373, \Lambda = 0.752$). Specifically, eccentric mean power ($p = 0.006, d = 0.686$), eccentric peak power ($p = 0.010, d = 0.654$), and eccentric peak velocity ($p = 0.027, d = 0.702$) experienced moderate-to-large increases between the first and the last testing sessions (Figure 2). However, despite the eccentric mean force

reaching the level of statistical significance between the first and the fourth testing time points, the observed increase was trivial in magnitude ($p = 0.053, d = 0.088$).

Discussion

The purpose of the present investigation was to examine season-long neuromuscular performance changes in female collegiate volleyball players. Our results reveal that the volleyball season is capable of inducing significant changes within the eccentric phase of the CVJ, whereas no alterations were observed during the concentric phase of the jumping motion. Specifically, eccentric mean power, peak power, and peak velocity increased from the initial to the final testing timepoints (i.e., ~15, 18, and 14% increase, respectively). Despite reaching the level of statistical significance, an increase in the eccentric mean force observed at the fourth testing time point when compared with the initial testing session was trivial in magnitude (i.e., ~2%). However, no changes were detected in the vertical jump height, RSI-modified, contraction time, and countermovement depth throughout the competitive season.

The findings of the present study pertaining to the concentric phase of the CVJ are contradictory to some of the previously published research reports (13,20,32). For example, Gannon et al. (13) examined changes in the neuromuscular performance of men's professional ice hockey players across a full competitive

Table 1
Changes in force-time metrics ($\bar{x} \pm SD$) across 5 different testing timepoints throughout a competitive season span.

Variable [unit]	Session 1	Session 2	Session 3	Session 4	Session 5
Eccentric phase					
Braking phase duration [s]	0.38 $\pm$ 0.08	0.37 $\pm$ 0.08	0.38 $\pm$ 0.07	0.37 $\pm$ 0.09	0.38 $\pm$ 0.09
Eccentric braking impulse [N·s]	33.6 $\pm$ 8.3	37.4 $\pm$ 10.2	36.5 $\pm$ 8.4	37.6 $\pm$ 7.4	39.0 $\pm$ 11.2
Eccentric duration [s]	0.61 $\pm$ 0.08	0.59 $\pm$ 0.08	0.60 $\pm$ 0.08	0.58 $\pm$ 0.09	0.59 $\pm$ 0.08
Eccentric peak velocity [m·s ⁻¹]	-0.92 $\pm$ 0.19	-0.99 $\pm$ 0.19	-0.96 $\pm$ 0.18	-1.02 $\pm$ 0.19	-1.05 $\pm$ 0.18†
Eccentric peak force [N]	1,430.3 $\pm$ 263.9	1,411.4 $\pm$ 247.8	1,388.4 $\pm$ 254.2	1,419.3 $\pm$ 224.5	1,426.8 $\pm$ 227.4
Eccentric mean force [N]	706.5 $\pm$ 127.8	706.5 $\pm$ 126.9	709.9 $\pm$ 125.7	717.4 $\pm$ 121.2†	716.8 $\pm$ 122.2
Eccentric peak power [W]	801.1 $\pm$ 223.3	875.3 $\pm$ 229.8	837.7 $\pm$ 213.5	908.8 $\pm$ 223.7	946.8 $\pm$ 222.6†
Eccentric mean power [W]	340.7 $\pm$ 75.4	361.2 $\pm$ 79.8	356.2 $\pm$ 76.9	381.5 $\pm$ 86.4	392.1 $\pm$ 74.5†
Concentric phase					
Concentric impulse [N·s]	159.2 $\pm$ 27.0	161.3 $\pm$ 27.8	156.9 $\pm$ 22.4	160.0 $\pm$ 21.9	159.7 $\pm$ 23.4
Concentric duration [s]	0.30 $\pm$ 0.04	0.31 $\pm$ 0.04	0.31 $\pm$ 0.04	0.31 $\pm$ 0.04	0.32 $\pm$ 0.05
Concentric peak velocity [m·s ⁻¹]	2.37 $\pm$ 0.17	2.40 $\pm$ 0.19	2.34 $\pm$ 0.15	2.36 $\pm$ 0.16	2.36 $\pm$ 0.16
Concentric peak force [N]	1,569.6 $\pm$ 267.4	1,530.8 $\pm$ 210.6	1,546.2 $\pm$ 250.0	1,539.4 $\pm$ 245.6	1,519.7 $\pm$ 231.6
Concentric mean force [N]	1,255.8 $\pm$ 210.9	1,230.3 $\pm$ 172.2	1,226.3 $\pm$ 185.1	1,239.1 $\pm$ 189.2	1,228.2 $\pm$ 188.7
Concentric peak power [W]	2,985.4 $\pm$ 527.2	2,990.2 $\pm$ 515.6	2,930.2 $\pm$ 433.2	2,994.3 $\pm$ 451.4	2,967.7 $\pm$ 444.4
Concentric mean power [W]	1,553.4 $\pm$ 282.9	1,529.6 $\pm$ 239.1	1,477.8 $\pm$ 221.1	1,509.4 $\pm$ 243.7	1,500.5 $\pm$ 241.9
Other					
Contraction time [s]	0.91 $\pm$ 0.11	0.90 $\pm$ 0.11	0.91 $\pm$ 0.95	0.89 $\pm$ 0.13	0.90 $\pm$ 0.11
Vertical jump height [cm]	25.3 $\pm$ 4.1	26.1 $\pm$ 4.9	24.5 $\pm$ 3.8	24.9 $\pm$ 4.1	24.9 $\pm$ 3.9
RSI-modified [m·s ⁻¹]	0.29 $\pm$ 0.06	0.30 $\pm$ 0.07	0.28 $\pm$ 0.06	0.30 $\pm$ 0.08	0.29 $\pm$ 0.08
Countermovement depth [cm]	-29.3 $\pm$ 4.8	-30.4 $\pm$ 4.6	-30.2 $\pm$ 2.1	-30.8 $\pm$ 6.3	-32.1 $\pm$ 4.9

RSI-modified = reactive strength index-modified.

†Statistically significant difference when compared with Session 1 ($p < 0.05$).

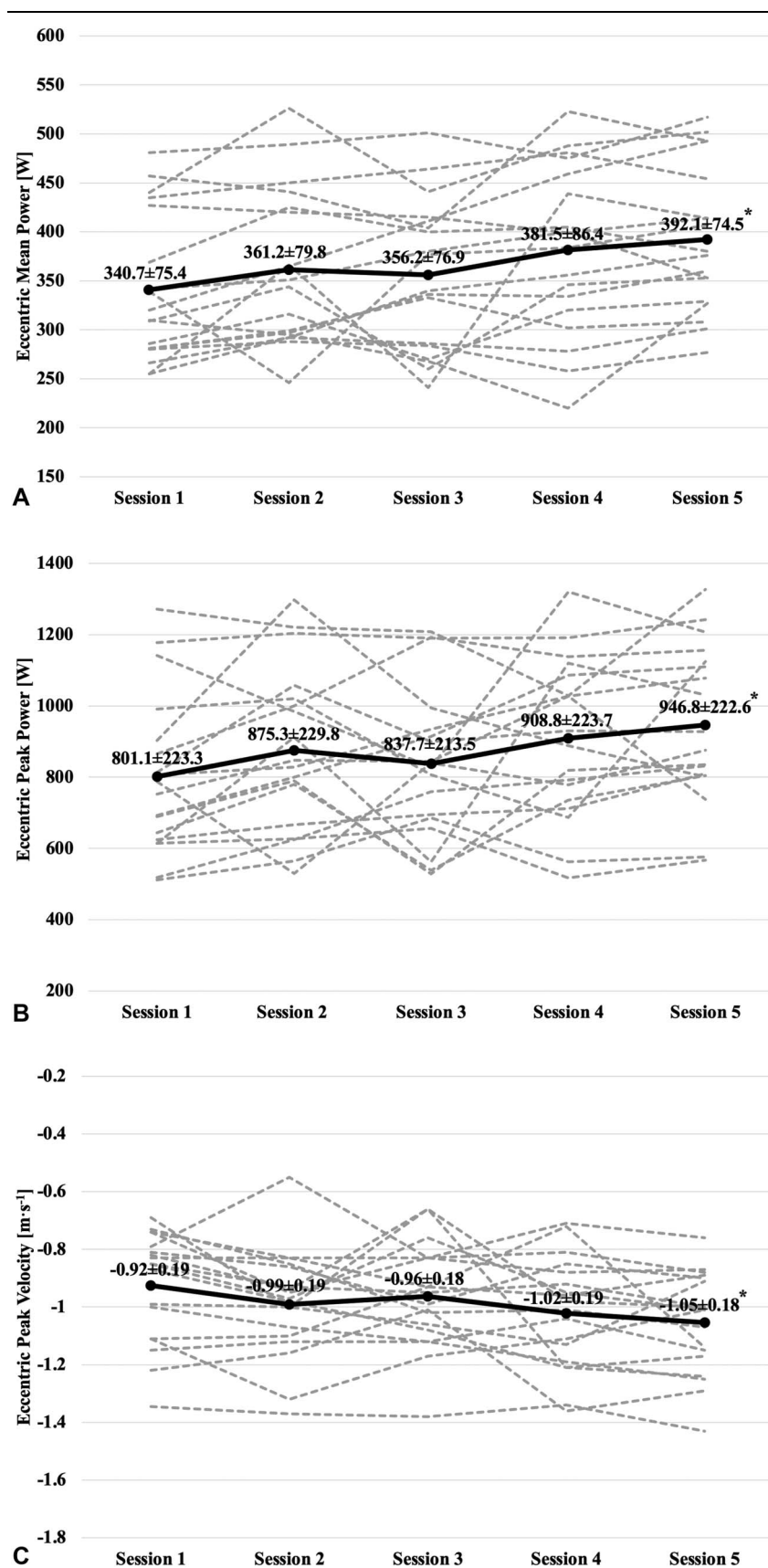


Figure 2. Changes in eccentric peak and mean power and peak velocity across 5 different testing sessions completed throughout a competitive season span. *Statistically significant difference when compared with session 1 ($p < 0.05$).

season. The authors found that concentric peak velocity drastically decreased from pre-season to the beginning of the competitive period (i.e., the first 9 weeks into the competition), with no differences being observed across other parts of the season (i.e., mid-season, late-season) (13). Similar observations were made by Spyrou et al. (32) when examining a cohort of professional male futsal players. A notable decrease in concentric peak power was found, whereas concentric peak velocity and duration tended to remain unchanged throughout the season (32). When compared with our results, these discrepancies in the findings may be largely attributed to the differences in the physiological demands of the sports (e.g., hockey, futsal, or volleyball), the level of competition (e.g., professional vs. collegiate), and athlete's sex (e.g., male vs. female). In addition, the data from our laboratory show that an intense training session in the duration of ~90 minutes had no impact on changes in force-time metrics during both concentric and eccentric phases of CVJ in professional male volleyball players (6), suggesting that highly trained athletes may be accustomed to dealing with high training volumes. In a similar manner, instead of assuming that the concentric force-time metrics analyzed in the present study lack sensitivity to detect neuromuscular fatigue, it would be more reasonable to consider that female collegiate volleyball players examined in this investigation were adequately prepared for their in-season competitive demands by their respective strength and conditioning staff. Although further research is warranted on this topic, the aforementioned speculation can be supported by the team's successful record despite a dense practice and competitive schedule, which included winning more than 20 games and being undefeated in conference play.

Although the majority of the previously conducted scientific literature has been primarily focused on analyzing the concentric phase of the CVJ, Merrigan et al. (27) in a recently published investigation suggested that the eccentric phase should not be overlooked because the phase-specific performance analysis approach may lead to the development of inadequate conclusions about an athlete's neuromuscular performance. Thus, along with the concentric phase of the CVJ, the present study examined the eccentric force-time characteristics and found that eccentric peak and mean power, and peak velocity significantly increased from the initial to the final testing timepoints. However, no significant changes were detected in the eccentric peak force, impulse, and duration through a competitive season span. So, we can assume that the observed increase in the athletes' eccentric peak power at the final testing timepoint could be primarily attributed to the increase in eccentric peak velocity (i.e., power = force $\times$ velocity) (12,31). As previously suggested, the observed increase in eccentric peak and mean power detected during the final testing timepoint could have occurred because of athletes having purposely reduced training loads (e.g., tapering phase) to optimize recovery before their conference tournament as the most important part of the competitive season (33). However, when solely focused on observing the regular-season competitive period (i.e., first-fourth testing timepoints), our results pertaining to the eccentric phase of the CVJ seem to align with the previously conducted research reports, indicating no changes in eccentric peak power and peak velocity across both futsal and basketball seasons (22,32). As previously mentioned, this may be largely attributed to the athlete's capability to sustain high-training and competitive loads. Although various factors such as the number of games played, frequency of the training sessions, and travel schedule could have contributed to this outcome, they were not examined in the present investigation. Thus, future research

should aim to include these factors along with CVJ monitoring to obtain a more comprehensive understanding of phase-specific changes in neuromuscular performance.

Finally, our results reveal that female collegiate volleyball athletes seem to attain similar outcome metrics (e.g., vertical jump height, RSI-modified) and movement strategies (e.g., counter-movement depth) throughout the competitive season span, which is in agreement with the previously published research reports (20,29,32). The magnitude of some of the previously mentioned performance parameters is similar to previous research and unpublished data collected in our laboratory focused on examining collegiate female volleyball players competing at the NCAA Division-I level of competition (e.g., vertical jump height = 27.9 ± 4.7 cm) (20). These findings further solidify the idea that coaches, sports scientists, and strength and conditioning practitioners should avoid solely focusing on monitoring a single CVJ metric but rather examine a broad spectrum of phase-specific variables that may allow for a deeper insight into the season-long neuromuscular fatigue changes and aid with planning and/or modifying the training regimens through the season (22,27).

Although providing valuable information regarding the impact of the competitive season on female volleyball athletes' neuromuscular fatigue, this study is not without limitations. The present investigation did not examine the internal and external training loads that athletes were exposed to during the season. Hence, future research should include workload monitoring strategies in addition to frequent neuromuscular performance assessments to obtain a more detailed understanding of performance fluctuations throughout the competitive season span. In addition, future research should examine if the findings obtained in the present study remain applicable to other levels of volleyball competition (e.g., professional) and if they are sex-specific.

Practical Applications

The findings of the present study indicate that both concentric and eccentric force-time metrics remain relatively unchanged throughout a regular season span within a cohort of highly trained collegiate female volleyball players. However, the eccentric metrics such as peak and mean power and peak velocity tend to display slight improvement after a brief tapering period. The aforementioned metrics may be the ones sensitive enough to identify the need for adjustments in training load to optimize recovery and ultimately improve on-court volleyball performance. However, the outcome metrics such as vertical jump height and RSI-modified did not display notable fluctuations across the season, suggesting that they may be less sensitive to detect neuromuscular performance changes throughout the competitive period. In addition, these findings can help coaches, sports scientists, and strength and conditioning practitioners to obtain a deeper insight into collegiate female athletes' force-time characteristics and their magnitudes and season-long fluctuations that may aid with developing adequate training regimens targeted toward optimizing athletes' performance.

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